

Erdős Problem #514

EP-514

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $f(z)$ be an entire transcendental function. Does there exist a path L so that, for every n , $|f(z)/z^n| \rightarrow \infty$ as $z \rightarrow \infty$ along L ?

Can the length of this path be estimated in terms of $M(r) = \max_{|z|=r} |f(z)|$? Does there exist a path along which $|f(z)|$ tends to ∞ faster than a fixed function of $M(r)$ (such that $M(r)^\epsilon$)?